

Modern Algebra

Assignment 7

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November 2025

6 Lagrange's Theorem

Problem 1

We observe that:

$$\begin{aligned}H &= 0 + H = 3 + H = \{-9, -6, -3, 0, 3, 6, 9\} \\1 + H &= \{-8, -5, -2, 1, 4, 7, 10\} \\2 + H &= \{-7, -4, -1, 2, 5, 8, 11\}\end{aligned}$$

Pick $t \in \mathbb{Z}$. Using the *Division Algorithm*, t can be written as $t = 3q + r$, where $(q \in \mathbb{Z}) \wedge (0 \leq r < 3)$. Clearly, when $(r = 0) \implies (t + H = H)$ as $3q \in H$. Moreover, when $r = 1, r = 2$, we see that

$$\begin{aligned}t + H &= (3q + 1) + H = 1 + (3q + H) = 1 + H \\t + H &= (3q + 2) + H = 2 + (3q + H) = 2 + H\end{aligned}$$

Problem 7

We know that $|\langle a \rangle| = |a| = 30$ and $|\langle a^4 \rangle| = \frac{30}{\gcd(30,4)} = 15$. Hence, $\frac{|\langle a \rangle|}{|\langle a^4 \rangle|} = \frac{30}{15} = 2$, so we have two cosets of $\langle a^4 \rangle$ in $\langle a \rangle$. Clearly, $a^0 \langle a^4 \rangle = a^6 \langle a^4 \rangle = \langle a^4 \rangle$ is a coset. The second coset is $a^5 \langle a^4 \rangle$.

Problem 9

Since the order of A_4 is $\frac{4!}{2} = 12$ and the order of H is 4, then there are $\frac{12}{4} = 3$ left cosets of H . The first coset is H itself, the other two are:

$$\begin{aligned}\alpha_5 H &= \{\alpha_5, \alpha_6, \alpha_7, \alpha_8\} \\ \alpha_9 H &= \{\alpha_9, \alpha_{10}, \alpha_{11}, \alpha_{12}\}\end{aligned}$$

For the other part, we know that $|S_4| = 4! = 24$. Thus, we can get $\frac{24}{4} = 6$ left cosets of H in S_4

Problem 10

Proof. Suppose H, K are subgroups from the group G , $a, b \in G$ and $aH = bK$. First, we pick an element $h \in H$. Since $aH = bK$, then there exist an element $k \in K$ such that $ah = bk$. We know that H is a subgroup, so it includes the identity element. So, $a \in aH$ and $a = bk'$ where $k' \in K$. We observe now that $ah = (bk')h = bk$. Multiplying by b^{-1} both sides and using the associative property, we get $(k')h = k \implies (k')^{-1}(k')h = (k')^{-1}k = h = (k')^{-1}k$ and $h \in K$ and $H \subset K$.

Next, we pick an element $k \in K$. Since $aH = bK$, then there exist an element $h \in H$ such that $ah = bk$. We know that K is a subgroup, so it includes the identity element. So, $b \in bK$ and $b = ah'$ where $h' \in H$. We observe now that $bh = (ah')h = ah$. Multiplying by a^{-1} both sides and using the associative property, we get $(h')h = h \implies (h')^{-1}(h')h = (h')^{-1}h = k = (h')^{-1}h$ and $k \in H$ and $K \subset H$. Therefore, since $H \subset K$ and $K \subset H$, then $H = K$, which satisfies the proof. ■

Problem 17

Proof. Suppose we have two primes p, q such that the order of group $G = pq$. Let H a proper subgroup of G , so the order of $H \neq pq$. By Lagrange's Theorem, $|H| \mid |G|$, so H is p, q or 1. Clearly, any group of order 1 is cyclic. Moreover, using *Corollary 7.2.6.* in Lecture Notes, since p, q are prime numbers, then any group of order equal to p or q is

cyclic. Therefore, we proved that any subgroup of the group G of order pq , where p, q are prime numbers, is a cyclic group. ■

Problem 19

We use *Fermat's Little Theorem*, for $7^{13} \mod 11$ we can see that 11 is a prime number. So, $7^{13} \mod 11 = ((7^{11} \mod 11) \times (7^2 \mod 11)) \mod 11 = (7 \times 5) \mod 11 = 2$. For $5^{15} \mod 7$, we can see that 7 is a prime number. So, $5^{15} \mod 7 = ((5^7 \mod 7) \times (5^7 \mod 7) \times (5 \mod 7)) \mod 7 = (5 \times 5 \times 5) \mod 7 = 6$.

Problem 22

Suppose H, K are subgroups of group G . We proved before in Assignemnt 3 that $H \cap K$ is a subgroup of both H, K . By *Lagrange's Theorem*, then $(|H \cap K| \mid |H|) \wedge (|H \cap K| \mid |K|)$. So, $|H \cap K| \mid \gcd(|K|, |H|)$. Thus, $|H \cap K|$ is a common divisor of both $|H|, |K|$. For the specific case we are given, we see that $\gcd(12, 35) = 1$ and $|H \cap K| = 1$.

Problem 38

Proof. Suppose G is a finite group. For the sake of contradiction, assume the index of $Z(G)$ is a prime. That is, there is a prime number of cosets of $Z(G)$ in G , call it p . So, $Z(G)$ is a proper subgroup and the group G is not abelian. The order of the quotient group $G/Z(G)$ is prime and the quotient group is cyclic. So, it can be generated using an element, call it $aZ(G)$. Now, every element in the quotient group can be written as a power of $aZ(G)$, let $k \in \mathbb{Z}$ be the power. So, $(aZ(G))^k = a^k Z(G)$. That shows that any $g \in G$ can be written as $g = a^k z$, where $z \in Z(G)$. Pick two elements $x, y \in G$ such that $x = a^i z_1, y = a^j z_2$, where $i, j \in \mathbb{Z} \wedge z_1, z_2 \in Z(G)$, and observe:

$$\begin{aligned} xy &= (a^i z_1)(a^j z_2) = (a^i (z_1 a^j) z_2) = (a^i a^j)(z_1 z_2) = a^{i+j}(z_1 z_2) \\ yx &= (a^j z_2)(a^i z_1) = (a^j (z_2 a^i) z_1) = (a^j a^i)(z_2 z_1) = a^{j+i}(z_2 z_1) = a^{i+j}(z_1 z_2). \end{aligned}$$

Thus, the group is abelian, which is a contradiction. Therefore, the index of $Z(G)$ is not prime. ■

Problem 39

Proof. The possible orders of $|H \cap K|$ are 1, 2, 4. If the order is 1, so it is the trivial group, which is abelian by definition. If the order is 2, 2 is prime, then the subgroup is cyclic and so it is abelian. Moreover, if the order is 4 we have two cases. First, if $|H \cap K|$

has an element of order 4, call it a . We observe that $\langle a^0, a^1, a^2, a^3 \rangle$ is the whole group $H \cap K$ and since it is cyclic, it is abelian. Second case when the group has no element of order 4, then all elements must be of order 2 except the identity. Observe that:

$$\begin{aligned}
 (ab)^2 &= e \\
 (ab)(ab) &= e \\
 a(ab)(ab) &= a \\
 (aa)b(ab) &= a \\
 b(ab) &= a \\
 b(ab)b &= ab \\
 ba(bb) &= ab \\
 ba &= ab.
 \end{aligned}$$

So, the group is abelian. Therefore, we proved that $H \cap K$ is an abelian group if $|H| = 24$ and $|K| \leq 20$. ■